Databricks ML Associate

High-Yield Next 100 (Set 3: Scenarios and Code Reading)

Practice Exam Deck

How to Use This Deck

100 more questions (Q1–Q100), complementing Sets 1 and 2, with more scenario, calculation, and code-reading items, on topics that commonly appear on the Databricks Machine Learning Associate exam.

Each question slide has four options (A–D); the next slide gives the correct answer and why the others are wrong.

Topic order: AutoML details; code reading; Feature Engineering scenarios; Spark DataFrame code; tuning scenarios; metric calculations; ML concepts; deployment, jobs, and monitoring; Unity Catalog governance.

Exam Domains Covered

Databricks Machine Learning: runtime, AutoML, MLflow, Unity Catalog registry, Feature Engineering client.

Data Processing: EDA, imputation, encoding, Spark ML transformers, pandas API on Spark and UDFs.

Model Development: Hyperopt, CrossValidator, metrics, bias/variance, leakage.

Model Deployment: batch, streaming, real-time serving, monitoring.

Question 1

Which AutoML argument lists columns that should not be used as features?

A. `skip_cols_list`
B. `drop_features`
C. `exclude_cols`
D. `ignore_rows`

Answer 1

Correct: C — it removes columns such as IDs from training.

Wrong:

A: not an argument.

B: not an argument.

D: not an argument.

Question 2

How do you tell AutoML how long it may run?

A. Set `max_evals`
B. Set `numFolds`
C. Set `autotermination_minutes` on the model
D. Set `timeout_minutes`

Answer 2

Correct: D — the experiment stops after the timeout.

Wrong:

A: that is a Hyperopt argument.

B: that is Spark ML CV.

C: that is a cluster setting.

Question 3

Which AutoML argument selects the metric used to rank trials?

A. `score_with`

B. `primary_metric`

C. `eval_fn`

D. `rank_by`

Answer 3

Correct: **B** — for example f1, roc_auc, rmse, or r2.

Wrong:

A: not an argument.

C: not an argument.

D: not an argument.

Question 4

Which argument makes AutoML split classification or regression data chronologically?

A. `split_seed`

B. `order_by`

C. `time_col`

D. `stratify_col`

Answer 4

Correct: C — it uses the earliest rows for training and later rows for validation and test.

Wrong:

A: not an argument.

B: not an argument.

D: not an argument.

Question 5

Which arguments are required for `databricks.automl.forecast()` beyond the data and target?

A. `exclude_cols` only

B. `time_col` and `horizon` (with a frequency)

C. `pos_label` and `threshold`

D. `numTrees` and `maxDepth`

Answer 5

Correct: B — forecasting needs the time axis and how far ahead to predict.

Wrong:

A: not enough for forecasting.

C: classification concepts.

D: tree hyperparameters.

Question 6

Which algorithm families does AutoML classification commonly try?

A. Prophet only

B. Transformers only

C. Decision trees, random forests, logistic regression, XGBoost, and LightGBM

D. k-means and PCA

Answer 6

Correct: C — AutoML tries several algorithm families.

Wrong:

A: a forecasting model.

B: not used for tabular AutoML here.

D: unsupervised methods.

Question 7

Where do AutoML trial notebooks and runs appear?

A. Only in the Spark UI
B. In a feature table
C. Only in the cluster logs
D. In an MLflow experiment

Answer 7

Correct: D — each trial is logged as a run with a notebook.

Wrong:

A: not stored there.

B: not stored there.

C: not stored there.

Question 8

A binary AutoML classification task has a rare positive class labelled "fraud". What does `pos_label` help with?

A. Excluding the class

B. Telling AutoML which class counts as positive for precision, recall, and F1

C. Choosing the cluster size

D. Setting the number of trees

Answer 8

Correct: B — metrics depend on which class is positive.

Wrong:

A: not its purpose.

C: not its purpose.

D: not its purpose.

Question 9

What does this code do? `with mlflow.start_run(): mlflow.log_param("depth", 4); mlflow.log_metric("rmse", 2.5)`

A. Creates two separate runs

B. Registers a model

C. Creates one run holding the parameter `depth` and the metric `rmse`

D. Raises an error because params and metrics cannot be mixed

Answer 9

Correct: C — both are logged to the same active run.

Wrong:

A: only one run is started.

B: no registration call.

D: they can.

Question 10

A notebook calls `mlflow.sklearn.log_model(model, "model")` outside any `start_run`. What happens in a Databricks notebook?

A. It always raises an error

B. It registers to Unity Catalog

C. It logs nothing

D. A run is started automatically and the model is logged to it

Answer 10

Correct: D — MLflow auto-creates a run when logging without one.

Wrong:

A: an active run is created.

B: no registered_model_name was given.

C: it logs the model.

Question 11

What does `mlflow.pyfunc.load_model("models:/main.ml.churn@Champion").predict(pdf)` return for a classifier logged with predicted labels?

A. The model signature only

B. The model's predictions for the rows in `pdf`

C. A new run ID

D. A registered model version

Answer 11

Correct: B — pyfunc models take a pandas DataFrame and return predictions.

Wrong:

A: not returned.

C: not returned.

D: not returned.

Question 12

A model was logged without a signature. What can happen when it is served?

A. It uses a GPU

B. It cannot be registered

C. It trains slower

D. Input schema validation is weaker, so mismatched columns may cause errors or silent issues

Answer 12

Correct: D — signatures help enforce expected inputs.

Wrong:

A: not related.

B: it can be registered in most cases.

C: not related.

Question 13

What is the result of `mlflow.search_runs(order_by=["metrics.f1 DESC"], max_results=1)` when searching experiment runs?

A. All runs

B. The most recent run always

C. The run with the highest F1

D. The run with the lowest F1

Answer 13

Correct: C — descending order puts the highest F1 first.

Wrong:

A: max_results limits to 1.

B: not sorted by time.

D: that would be ASC.

Question 14

Which line is correct for logging a Spark ML model?

A. `mlflow.sklearn.log_model(pipeline_model, "model")`

B. `mlflow.spark.save(pipeline_model)`

C. `mlflow.spark.log_model(pipeline_model, "model")`

D. `mlflow.log_dataframe(pipeline_model)`

Answer 14

Correct: C — use the Spark flavor for Spark ML models.

Wrong:

A: wrong flavor.

B: not the logging API.

D: not an API.

Question 15

Which code builds a Spark ML pipeline correctly?

A. `Pipeline(indexer, encoder, assembler, rf)`

B. `Pipeline(stages=rf, assembler)`

C. `Pipeline.add(indexer).add(rf)`

D. `Pipeline(stages=[indexer, encoder, assembler, rf])`

Answer 15

Correct: D — stages are listed in order.

Wrong:

A: stages must be passed as a list.

B: invalid syntax.

C: not the API.

Question 16

What does this return? `rf = RandomForestClassifier(featuresCol="features", labelCol="label"); model = rf.fit(train)`

A. A fitted `RandomForestClassificationModel`

B. A predictions DataFrame

C. A `Pipeline`

D. A metric value

Answer 16

Correct: A — fit on an estimator returns a model.

Wrong:

B: that is `transform`.

C: not returned.

D: not returned.

Question 17

What does `model.transform(test)` return for a Spark ML classifier?

A. A DataFrame with added columns such as `prediction` and `probability`

B. A list of weights

C. A metric

D. A new model

Answer 17

Correct: A — transform adds output columns.

Wrong:

B: not returned.

C: use an evaluator for that.

D: that is fit.

Question 18

Which code evaluates AUC for a Spark ML model's predictions?

A. `model.score(preds)`

B. `BinaryClassificationEvaluator(metricName="areaUnderROC").evaluate(preds)`

C. `RegressionEvaluator(metricName="areaUnderROC").evaluate(preds)`

D. `preds.auc()`

Answer 18

Correct: B — the evaluator takes the predictions DataFrame.

Wrong:

A: not the Spark ML API.

C: wrong evaluator.

D: not an API.

Question 19

Which Hyperopt call is correct for minimizing `objective` over `space` with TPE for 20 trials?

A. `tpe.fmin(space)`

B. `fmin(fn=objective, space=space, algo=tpe.suggest, max_evals=20)`

C. `minimize(objective, space, 20)`

D. `fmin(objective, space, algo=tpe, max_evals=20)`

Answer 19

Correct: B — fn, space, algo, and max_evals are the core arguments.

Wrong:

A: not the API.

C: not the API.

D: algo needs `tpe.suggest`.

Question 20

An objective returns `-accuracy` while the team reports the best result. What should they report?

A. The number of trials
B. The loss as the accuracy directly
C. The worst loss
D. The negated best loss, which is the best accuracy

Answer 20

Correct: D — loss is -accuracy, so flip the sign.

Wrong:

A: not a metric.

B: the sign is flipped.

C: not the best.

Question 21

A scikit-learn Hyperopt objective is run with `SparkTrials(parallelism=8)` on an 8-worker cluster. What is the main benefit?

A. Up to 8 trials can run at the same time on workers

B. TPE becomes random search

C. Each model trains on all workers

D. Trials are guaranteed to be better

Answer 21

Correct: A — SparkTrials distributes single-node trials.

Wrong:

B: TPE still runs.

C: sklearn does not distribute.

D: not guaranteed.

Question 22

With SparkTrials, how are MLflow runs for trials typically organized?

A. One run per cluster node
B. A parent run with a child run per trial
C. One run per Spark task
D. No runs are created

Answer 22

Correct: B — Hyperopt on Databricks logs trials as nested runs.

Wrong:

A: not how it is organized.

C: not how it is organized.

D: runs are created.

Question 23

Which snippet pulls the Python value of the best run's RMSE from a `search_runs` DataFrame `runs`?

A. `runs.get_metric()`

B. `runs.rmse()`

C. `runs["rmse"].best`

D. `runs.loc[0, "metrics.rmse"]`

Answer 23

Correct: D — metrics columns are named `metrics.<name>`.

Wrong:

- A: not valid.
- B: not valid.
- C: not valid.

Question 24

What does `df.randomSplit([0.7, 0.3], seed=1)` return?

A. A dict of metrics
B. A list of two DataFrames
C. A Pipeline
D. A single DataFrame

Answer 24

Correct: B — the weights define the split.

Wrong:

A: not returned.

C: not returned.

D: it returns multiple.

Question 25

Which code removes the `id` column and keeps the rest?

A. `df.select("id")`

B. `df.drop("id")`

C. `df.remove("id")`

D. `df.filter("id")`

Answer 25

Correct: B — drop removes the named column.

Wrong:

A: keeps only id.

C: not an API.

D: filters rows.

Question 26

Which code keeps rows where age is greater than 30?

A. `df.limit(df.age > 30)`

B. `df.select(df.age > 30)`

C. `df.drop(df.age > 30)`

D. `df.filter(df.age > 30)`

Answer 26

Correct: D — filter keeps rows matching the condition.

Wrong:

A: limit takes a number.

B: returns a boolean column.

C: drops columns.

Question 27

Which code casts `price` to double?

A. `df.withColumn("price", df.price.cast("double"))`

B. `df.price.toDouble()`

C. `df.drop("price", "double")`

D. `df.convert("price", "double")`

Answer 27

Correct: A — cast changes the column type.

Wrong:

B: not an API.

C: drops columns.

D: not an API.

Question 28

Which code creates a Spark DataFrame from a pandas DataFrame `pdf`?

A. `spark.createDataFrame(pdf)`

B. `pdf.toSpark()`

C. `spark.load(pdf)`

D. `pdf.to_delta()`

Answer 28

Correct: A — it converts pandas to Spark.

Wrong:

B: not an API.

C: not an API.

D: not the conversion.

Question 29

Training features were computed with a window using future data relative to each label date. What is the problem?

A. Slow inference

B. Underfitting only

C. Too few trees

D. Data leakage that makes offline metrics too optimistic

Answer 29

Correct: D — the model sees information unavailable at prediction time.

Wrong:

A: not the problem.

B: not the main problem.

C: not the problem.

Question 30

A feature table has `customer_id` as key and a `ts` timestamp key. The label table has `customer_id` and `event_time`. What is needed for a point-in-time join?

A. `lookup_key="ts"` only

B. A `FeatureLookup` with `timestamp_lookup_key="event_time"`

C. A cross join

D. Rounding all dates

Answer 30

Correct: B — this joins the latest feature values at or before each event time.

Wrong:

A: ignores the label time column.

C: causes leakage and row explosion.

D: does not fix joins.

Question 31

A team wants to add a new column to an existing feature table without overwriting existing rows. What should they do?

A. Create a new cluster

B. Drop and recreate the table every time

C. Use `mode="overwrite"` with only new rows

D. Write the new column with `fe.write_table(..., mode="merge")` keyed on the primary key

Answer 31

Correct: D — merge adds or updates by key.

Wrong:

A: not related.

B: loses history.

C: replaces data.

Question 32

Training uses `FeatureLookups` over two feature tables. Which column must exist in the label DataFrame?

A. The lookup key columns (and a timestamp column for point-in-time lookups)

B. A copy of every feature

C. Only the label column

D. An MLflow run ID

Answer 32

Correct: A — keys connect the label rows to feature rows.

Wrong:

B: unnecessary.

C: keys are required too.

D: not needed.

Question 33

Why might offline model accuracy be good but online accuracy poor?

A. The cluster is too small

B. The model has too few parameters

C. Features are computed differently online than offline (training-serving skew)

D. MLflow is not installed

Answer 33

Correct: C — a feature store reduces this risk.

Wrong:

A: not implied.

B: not implied.

D: not implied.

Question 34

Which key type does `fe.create_table` need so that rows can be uniquely identified?

A. `foreign_keys`
B. `cluster_keys` only
C. `primary_keys`
D. `partition_keys` only

Answer 34

Correct: C — primary keys identify feature rows.

Wrong:

A: not required.

B: for layout only.

D: for layout only.

Question 35

A feature table is large and only some features are needed for a model. How do you limit the features used?

A. Drop the table

B. Use `df.limit(1)`

C. Specify `feature_names` in the `FeatureLookup`

D. Disable lineage

Answer 35

Correct: C — only the listed features are joined.

Wrong:

A: loses features.

B: reduces rows, not features.

D: unrelated.

Question 36

What is true about a model logged by `fe.log_model()` when scored with `fe.score_batch()`?

A. All features must be in the input DataFrame

B. The model is retrained

C. Feature values are looked up from the feature tables by key

D. Scoring always uses the training data

Answer 36

Correct: C — the input DataFrame only needs keys and non-lookup inputs.

Wrong:

A: that defeats the lookup.

B: not true.

D: not true.

Question 37

Which code adds a column `is_high` that is 1 when `amount > 100` and 0 otherwise?

A. `df.withColumn("is_high", F.when(df.amount > 100, 1).otherwise(0))`

B. `df.withColumn("is_high").when(1)`

C. `df.add("is_high", df.amount > 100)`

D. `df.if("amount", 100)`

Answer 37

Correct: A — when/otherwise builds conditional columns.

Wrong:

B: invalid.

C: not an API.

D: not an API.

Question 38

Which aggregation computes the average `price` per `region`?

A. `df.avg("region", "price")`

B. `df.agg("region").avg("price")`

C. `df.groupBy("price").agg(F.avg("region"))`

D. `df.groupBy("region").agg(F.avg("price"))`

Answer 38

Correct: D — groupBy then agg.

Wrong:

A: not an API.

B: invalid.

C: swapped columns.

Question 39

Which join keeps all rows from the left DataFrame, with nulls where the right has no match?

A. Left join
B. Inner join
C. Cross join
D. Left anti join

Answer 39

Correct: A — left join preserves all left rows.

Wrong:

B: keeps only matches.

C: returns all pairs.

D: keeps only non-matching rows.

Question 40

Which join finds left rows with no match in the right DataFrame?

A. Left anti join
B. Inner join
C. Full outer join
D. Left semi join

Answer 40

Correct: A — it returns non-matching left rows.

Wrong:

B: returns matches.

C: returns all rows.

D: returns matching left rows.

Question 41

Which call stacks two DataFrames with the same schema vertically?

A. `df1.union(df2)`

B. `df1.stack(df2)`

C. `df1.join(df2)`

D. `df1.merge(df2)`

Answer 41

Correct: A — union appends rows.

Wrong:

B: not an API.

C: combines columns by key.

D: not a Spark DataFrame method.

Question 42

Which code orders a Spark DataFrame by `score` highest first?

A. `df.limit("score")`

B. `df.sortUp("score")`

C. `df.orderBy(F.col("score").desc())`

D. `df.orderBy("score", asc=True)`

Answer 42

Correct: C — desc sorts in descending order.

Wrong:

A: limit takes a count.

B: not an API.

D: ascending.

Question 43

A pandas-on-Spark DataFrame `psdf` is needed as a Spark DataFrame again. Which call converts it?

A. `spark.from_pandas(psdf)`

B. `psdf.collect_spark()`

C. `psdf.to_spark()`

D. `psdf.toDF()`

Answer 43

Correct: C — to_spark converts to a Spark DataFrame.

Wrong:

A: not an API.

B: not an API.

D: not the pandas API on Spark method.

Question 44

A scikit-learn model must score a very large Spark DataFrame. Which approach distributes scoring?

A. `df.coalesce(1)` and loop

B. `df.cache()` only

C. A pandas UDF or `mlflow.pyfunc.spark_udf` wrapping the model

D. `df.toPandas().predict()`

Answer 44

Correct: C — both run predictions on workers.

Wrong:

A: not distributed.

B: does not score.

D: runs on the driver only.

Question 45

Which Spark SQL statement creates a table from a query result?

A. `CREATE TABLE t AS SELECT ...`

B. `MAKE TABLE t FROM ...`

C. `COPY TABLE t`

D. `SELECT INTO t`

Answer 45

Correct: A — CTAS creates a table from a query.

Wrong:

B: not valid.

C: not valid.

D: not Spark SQL.

Question 46

What does `display(df)` offer in a Databricks notebook?

A. A deployed endpoint

B. A trained model

C. A Git commit

D. An interactive table with built-in charting options

Answer 46

Correct: D — it renders data with plots.

Wrong:

A: not provided.

B: not provided.

C: not provided.

Question 47

A 5-fold CrossValidator uses a grid of 2 x 3 x 2 combinations. How many models are trained during the search (excluding the final refit)?

A. 60

B. 12

C. 30

D. 17

Answer 47

Correct: A — 12 combinations times 5 folds.

Wrong:

B: ignores the folds.

C: miscounts the grid or folds.

D: adds instead of multiplying.

Question 48

A team has limited compute and many hyperparameters. Which strategy is usually more efficient than full grid search?

A. Removing validation

B. Random search or Bayesian optimization (Hyperopt/TPE)

C. Training on the test set

D. Exhaustive grid with finer steps

Answer 48

Correct: B — they explore the space with fewer trials.

Wrong:

A: invalid.

C: invalid.

D: more costly.

Question 49

A tuning job finds the best validation score at the boundary of the search range. What should the team do?

A. Expand the search range in that direction and rerun

B. Accept it as globally optimal

C. Shrink the range

D. Switch to the test set

Answer 49

Correct: A — the optimum may lie outside the range.

Wrong:

B: may be wrong.

C: makes it worse.

D: invalid.

Question 50

Why is the test set not used for hyperparameter tuning?

A. Using it would bias the final performance estimate
B. MLflow forbids it
C. It is too small to compute metrics
D. Hyperopt cannot read it

Answer 50

Correct: A — tuning on test data leaks information.

Wrong:

B: not the reason.

C: not the reason.

D: not the reason.

Question 51

What does `TrainValidationSplit` need compared with `CrossValidator`?

A. A Delta trigger
B. A serving endpoint
C. A feature table
D. A `trainRatio` instead of `numFolds`

Answer 51

Correct: D — trainRatio sets the split proportion.

Wrong:

A: not required.

B: not required.

C: not required.

Question 52

A tuned model performs well in CV but poorly on the held-out test set. What is a likely cause?

A. The test set is perfect
B. MLflow logged too many metrics
C. Too many workers
D. Overfitting to the validation process or leakage during CV

Answer 52

Correct: D — tuning can overfit validation folds.

Wrong:

A: not implied.

B: not related.

C: not related.

Question 53

Where should a Spark ML `Pipeline` containing preprocessing be placed when tuning with `CrossValidator`?

A. Run only on the test data

B. As the estimator inside the CrossValidator so preprocessing is refit per fold

C. Skip preprocessing

D. Fit once on all data before CV

Answer 53

Correct: **B** — this avoids leakage across folds.

Wrong:

A: invalid.

C: not required.

D: leaks fold information.

Question 54

Hyperopt trials all return the same loss. What is a likely cause?

A. The cluster is too large
B. The dataset is too small
C. TPE is too random
D. The objective ignores the sampled hyperparameters

Answer 54

Correct: D — the parameters are not passed into the model.

Wrong:

A: not related.

B: not necessarily related.

C: does not explain identical loss.

Question 55

A classifier has TP=40, FP=10, FN=20, TN=30. What is its precision?

A. 0.67

B. 0.70

C. 0.50

D. 0.80

Answer 55

Correct: D — 40 / (40 + 10).

Wrong:

A: that is recall, 40/60.

B: that is accuracy, 70/100.

C: incorrect.

Question 56

A classifier has TP=40, FP=10, FN=20, TN=30. What is its recall?

A. 0.50

B. 0.30

C. 0.80

D. About 0.67

Answer 56

Correct: D — 40 / (40 + 20).

Wrong:

A: incorrect.

B: incorrect.

C: that is precision.

Question 57

A classifier has TP=40, FP=10, FN=20, TN=30. What is its accuracy?

A. 0.67

B. 0.80

C. 0.40

D. 0.70

Answer 57

Correct: D — (40 + 30) / 100.

Wrong:

A: that is recall.

B: that is precision.

C: incorrect.

Question 58

Prediction errors are 1, 1, 1, and 5. What is the MAE?

A. 2.0

B. 8.0

C. 2.65

D. 1.0

Answer 58

Correct: A — $(1 + 1 + 1 + 5) / 4$.

Wrong:

B: that is the sum.

C: that is the RMSE.

D: ignores the large error.

Question 59

Prediction errors are 1, 1, 1, and 5. Which is larger, RMSE or MAE, and why?

A. RMSE, because squaring weights the large error more

B. MAE, because it sums errors

C. Neither can be computed

D. They are always equal

Answer 59

Correct: A — RMSE is about 2.65 versus MAE 2.0.

Wrong:

B: MAE averages absolute errors.

C: both can be computed.

D: they differ.

Question 60

A bank wants to minimize wrongly flagging good customers as fraud. Which metric focuses on this?

A. Recall

B. R2

C. Precision

D. Silhouette

Answer 60

Correct: C — precision penalizes false positives.

Wrong:

A: focuses on false negatives.

B: regression metric.

D: clustering metric.

Question 61

A ranking-quality metric that is threshold independent for binary classification is:

A. ROC AUC

B. Precision at 0.5

C. Accuracy at 0.5

D. F1 at 0.5

Answer 61

Correct: A — it summarizes performance over all thresholds.

Wrong:

B: threshold dependent.

C: threshold dependent.

D: threshold dependent.

Question 62

For highly imbalanced data, which curve is often more informative than the ROC curve?

A. The residual plot

B. The precision-recall curve

C. The learning-rate curve

D. The elbow curve

Answer 62

Correct: B — it focuses on the positive class.

Wrong:

A: for regression.

C: not a standard evaluation.

D: for clustering.

Question 63

When averaging F1 across classes in multiclass classification, what does macro averaging do?

A. Pools all predictions first
B. Ignores rare classes
C. Computes F1 per class and averages them equally
D. Weights classes by frequency

Answer 63

Correct: C — each class counts the same.

Wrong:

A: that is micro averaging.

B: not what it does.

D: that is weighted averaging.

Question 64

A model with R2 = -0.2 on the test set indicates:

A. No variance in the labels

B. It is worse than predicting the mean

C. A perfect fit

D. A bug in the evaluator only

Answer 64

Correct: B — R2 can be negative on held-out data.

Wrong:

A: not implied.

C: that is near 1.

D: not necessarily.

Question 65

Which is an example of supervised learning?

A. Reducing dimensions with PCA
B. Grouping customers without labels
C. Detecting topics without labels
D. Predicting house prices from labelled sales data

Answer 65

Correct: D — labels guide the training.

Wrong:

A: unsupervised.

B: unsupervised.

C: unsupervised.

Question 66

Which is an example of unsupervised learning?

A. Classifying spam from labelled email
B. Clustering customers into segments without labels
C. Forecasting sales from history
D. Predicting churn from labels

Answer 66

Correct: B — no target is used.

Wrong:

A: supervised.

C: supervised.

D: supervised.

Question 67

Which algorithm classifies a point by the majority label of its nearest neighbors?

A. k-means
B. k-nearest neighbors
C. PCA
D. Linear regression

Answer 67

Correct: B — it uses distance to neighbors.

Wrong:

A: unsupervised clustering.

C: dimensionality reduction.

D: predicts continuous values.

Question 68

Why does k-nearest neighbors usually need feature scaling?

A. Distances are dominated by features with large numeric ranges

B. It does not use distances

C. It uses only categorical features

D. It cannot handle numbers

Answer 68

Correct: A — scaling puts features on comparable scales.

Wrong:

B: it does.

C: not true.

D: it can.

Question 69

What is a main advantage of a random forest over a single decision tree?

A. It needs no data
B. Lower variance and less overfitting through averaging many trees
C. It never overfits
D. It is fully interpretable

Answer 69

Correct: B — bagging reduces variance.

Wrong:

A: not true.

C: it can still overfit.

D: a single tree is more interpretable.

Question 70

In gradient boosting, how are trees combined?

A. In parallel as independent trees only
B. Sequentially, each tree fitting the errors of the previous ensemble
C. By clustering features
D. By averaging identical trees

Answer 70

Correct: B — boosting corrects residual errors step by step.

Wrong:

A: that is bagging.

C: not boosting.

D: not boosting.

Question 71

Which scenario describes underfitting?

A. Perfect test accuracy
B. Training stops early by design
C. Low accuracy on both training and validation data
D. High training accuracy, low validation accuracy

Answer 71

Correct: C — the model is too simple to learn the pattern.

Wrong:

A: not underfitting.

B: not underfitting.

D: that is overfitting.

Question 72

Which action most directly reduces overfitting?

A. Train on the test set

B. Remove the validation set

C. Add regularization or use more training data

D. Increase model complexity

Answer 72

Correct: C — both reduce variance.

Wrong:

A: invalid.

B: removes the signal.

D: can increase overfitting.

Question 73

Which action can reduce underfitting?

A. Add stronger regularization

B. Use a more flexible model or add informative features

C. Shuffle labels

D. Reduce training data

Answer 73

Correct: B — it increases the model's capacity.

Wrong:

A: can worsen underfitting.

C: destroys signal.

D: does not help.

Question 74

Why is stratified k-fold useful for classification?

A. It speeds up training by ignoring labels
B. It removes the minority class
C. It shuffles labels
D. Each fold keeps roughly the same class proportions

Answer 74

Correct: D — this stabilizes evaluation on imbalanced data.

Wrong:

A: it does not.

B: it does not.

C: it does not.

Question 75

A feature is the label's exact copy under a different name. What is this called?

A. Cross-validation
B. Regularization
C. Target leakage
D. Normalization

Answer 75

Correct: C — the feature reveals the answer.

Wrong:

A: not leakage.

B: not leakage.

D: not leakage.

Question 76

Logistic regression outputs probabilities. How do you control the trade-off between precision and recall?

A. Adjust the decision threshold

B. Change the cluster size

C. Change the Delta table version

D. Change the Git branch

Answer 76

Correct: A — a higher threshold typically raises precision and lowers recall.

Wrong:

B: unrelated.

C: unrelated.

D: unrelated.

Question 77

A batch scoring job needs to write predictions to Delta daily. Which pattern is typical?

A. Call a REST endpoint per row from the driver

B. Email predictions

C. Read input from Delta, apply the model with `spark_udf` or `score_batch`, then write results to a Delta table

D. Save predictions in notebook variables

Answer 77

Correct: C — it is a standard batch inference pipeline.

Wrong:

A: does not scale.

B: not a pipeline.

D: not durable.

Question 78

Why might a team prefer batch inference over real-time serving?

A. It does not require a model

B. It always has lower latency

C. It avoids data entirely

D. Predictions are needed periodically and cost and throughput matter more than latency

Answer 78

Correct: D — batch is cheaper for large volumes.

Wrong:

A: it does.

B: it has higher latency.

C: it uses data.

Question 79

A real-time use case needs sub-second predictions. Which option fits best?

A. A notebook run manually
B. A Model Serving endpoint
C. A Delta `VACUUM`
D. A nightly job

Answer 79

Correct: B — endpoints provide low-latency REST inference.

Wrong:

A: not production-grade.

C: unrelated.

D: too slow.

Question 80

Which workload setting on a serving endpoint affects how many concurrent requests it can handle?

A. The notebook theme
B. Workload size (concurrency)
C. The Git branch
D. The MLflow experiment name

Answer 80

Correct: B — larger sizes handle more concurrent requests.

Wrong:

A: unrelated.

C: unrelated.

D: unrelated.

Question 81

A team wants to compare a new model against the current one using live traffic safely. What is a good approach?

A. Route a small share of traffic to the new version on the same endpoint

B. Replace the production model without testing

C. Test only on the training data

D. Disable monitoring

Answer 81

Correct: A — traffic splitting supports gradual rollout.

Wrong:

B: risky.

C: not safe.

D: removes safety.

Question 82

A new model version performs badly in production. What is the fastest rollback when clients load `@Champion`?

A. Delete the new model's experiment

B. Rewrite all clients

C. Retrain from scratch immediately

D. Point the `Champion` alias back to the previous version

Answer 82

Correct: D — clients pick up the older version.

Wrong:

A: does not roll back.

B: unnecessary.

C: slower.

Question 83

What should a CI/CD process for models typically include before promotion?

A. Manual copying of files between workspaces only
B. Automated tests and validation against baseline metrics
C. Skipping evaluation
D. Deleting old versions first

Answer 83

Correct: B — promotion should be gated by checks.

Wrong:

A: error prone.

C: risky.

D: prevents rollback.

Question 84

Which job setting reruns a task automatically after a transient failure?

A. A cluster policy
B. Auto-termination
C. Task retries
D. A model alias

Answer 84

Correct: C — retries help with temporary errors.

Wrong:

A: limits configuration.

B: a cluster setting.

D: labels versions.

Question 85

How do you pass a parameter to a notebook task in a job and read it in the notebook?

A. Use `df.cache()`
B. Edit the notebook for each run
C. Define a job parameter and read it with `dbutils.widgets.get("name")`
D. Use `dbutils.fs.put`

Answer 85

Correct: C — job parameters appear as widgets.

Wrong:

A: not for parameters.

B: unnecessary.

D: writes files.

Question 86

Which tool captures serving request/response data to help monitor model quality?

A. Git folders

B. Inference tables

C. Cluster policies

D. `ZORDER`

Answer 86

Correct: B — payloads are logged to a Delta table.

Wrong:

A: version control.

C: limit configuration.

D: file layout.

Question 87

Ground-truth labels arrive weeks after predictions. How can you monitor model quality in the meantime?

A. Drop the monitoring tables

B. Retrain every minute

C. Track input drift and prediction distributions, then compute accuracy once labels arrive

D. Do nothing until labels arrive

Answer 87

Correct: C — proxy signals detect issues earlier.

Wrong:

A: removes visibility.

B: wasteful.

D: misses early problems.

Question 88

Which change in the input data indicates data drift?

A. The experiment name changes
B. The model file size grows
C. The cluster restarts
D. The distribution of a feature in production differs from training

Answer 88

Correct: D — inputs have shifted.

Wrong:

A: not drift.

B: not drift.

C: not drift.

Question 89

Why log the training dataset version or Delta table version with each run?

A. To reproduce the exact data used to train the model

B. To avoid needing a signature

C. To reduce model size

D. To speed up training

Answer 89

Correct: A — reproducibility requires data lineage.

Wrong:

B: not the purpose.

C: not the purpose.

D: not the purpose.

Question 90

Which element makes an MLflow model reproducible on another machine?

A. Only the run name

B. The cluster ID

C. Logged dependencies (requirements or conda environment) along with the model artifact

D. Only the notebook theme

Answer 90

Correct: C — the environment can be recreated.

Wrong:

A: not enough.

B: not portable.

D: not related.

Question 91

Which Unity Catalog object is the top-level container below the metastore?

A. Schema
B. Table
C. Volume
D. Catalog

Answer 91

Correct: D — the hierarchy is metastore, catalog, schema, then tables, volumes, models.

Wrong:

A: sits under a catalog.

B: sits under a schema.

C: sits under a schema.

Question 92

What does `GRANT USE SCHEMA ON SCHEMA main.ml TO group` allow, together with other needed privileges?

A. Creating clusters

B. Access to objects within the schema

C. Reading every table automatically

D. Deleting the catalog

Answer 92

Correct: **B** — USE SCHEMA is required to reach objects inside it.

Wrong:

A: not granted.

C: SELECT is still needed.

D: not granted.

Question 93

Who owns a newly created Unity Catalog table by default?

A. The cluster

B. The metastore only

C. The user or principal who created it

D. All workspace users

Answer 93

Correct: C — owners can manage grants on the object.

Wrong:

A: not an owner.

B: not the default owner.

D: not by default.

Question 94

Why use a service principal to run a production job?

A. It removes the need for permissions

B. It avoids tying the job to an individual user's account

C. It replaces Unity Catalog

D. It makes models more accurate

Answer 94

Correct: B — service principals suit automation.

Wrong:

A: it still needs permissions.

C: it does not.

D: not related.

Question 95

A column contains sensitive personal data. Which Unity Catalog capability can restrict who sees raw values?

A. Column masks or row filters

B. `mlflow.autolog()`

C. `OPTIMIZE`

D. `df.cache()`

Answer 95

Correct: A — they apply fine-grained access control.

Wrong:

B: logs runs.

C: compacts files.

D: does not control access.

Question 96

What is the benefit of governing models and features in Unity Catalog?

A. Automatic accuracy gains

B. Removal of the need for validation

C. Centralized access control, auditing, and lineage across data and models

D. Free compute

Answer 96

Correct: C — one governance layer covers data and ML assets.

Wrong:

A: not provided.

B: not provided.

D: not provided.

Question 97

Which code fills missing values in column age with the median in Spark ML?

A. `Imputer(inputCols=["age"], outputCols=["age_imp"], strategy="median").fit(df).transform(df)`

B. `Imputer(strategy="median").transform(df)`

C. `df.fillna(median)` with no value computed

D. `StandardScaler(strategy="median")`

Answer 97

Correct: A — Imputer is fit and then used to transform.

Wrong:

B: must fit first and set columns.

C: no median is computed.

D: not a StandardScaler option.

Question 98

What input does Spark ML `StandardScaler` require?

A. A pandas Series

B. A vector column, for example from `VectorAssembler`

C. A Delta table path

D. A string column

Answer 98

Correct: B — scalers operate on vector columns.

Wrong:

A: not supported.

C: not supported.

D: not supported.

Question 99

Which call sets the MLflow experiment by workspace path?

A. `mlflow.start_path(path)`

B. `mlflow.log_experiment(path)`

C. `mlflow.use_experiment(path)`

D. `mlflow.set_experiment("/Users/me@company.com/churn")`

Answer 99

Correct: D — it creates or selects the experiment at the path.

Wrong:

A: not an API.

B: not an API.

C: not an API.

Question 100

Which workload most benefits from a GPU cluster?

A. Running `OPTIMIZE` on Delta
B. Training a small scikit-learn linear model
C. Writing a CSV file
D. Training a deep neural network

Answer 100

Correct: D — GPUs accelerate large matrix operations.

Wrong:

A: does not need GPUs.

B: CPU is enough.

C: does not need GPUs.